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DEPARTMENT OF TEACHER EDUCATION
SCHOOL OF EDUCATION AND LEADERSHIP
COLLEGES OF EDUCATION
END OF YEAR THREE SEMESTER ONE EXAMINATIONS, 2023/2024
B. ED. PROGRAMME

COURSE CODE: TEJS 305

COURSE TITLE: DATABASE SYSTEMS AND SOFTWARE DEVELOPMENT

Instruction: Answer **all** questions in Section A and any **three** in Section B.

Time: 2 hours

SECTION A

Answer all the questions in this section. [25 Marks]

1. Which of the following best describes a DBMS?
   a) A software system used to create and manage databases
   b) A collection of files stored on a computer
   c) A network of interconnected databases
   d) A physical storage device for data

2. What does SQL stand for in the context of DBMS?
   a) Structured Query Language
   b) System Query Language
   c) Standardized Query Logic
   d) Sequential Query Language

3. What is a database?
   a) Organized collection of information that cannot be accessed, updated, and managed
   b) Collection of data or information without organizing
   c) Organized collection of data or information that can be accessed, updated, and managed
   d) Organized collection of data that cannot be updated

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4. Which of the following is not a type of DBMS architecture?
   a) Hierarchical
   b) Network
   c) Relational
   d) Linear

5. Related fields in a data base are grouped to form.....
   a) Data file
   b) Data record
   c) Menu
   d) Bank

6. What is DBMS?
   a) DBMS is a collection of queries
   b) DBMS is a high-level language
   c) DBMS is a programming language
   d) DBMS stores, modifies and retrieves data

7. Which type of data can be stored in the database?
   a) Image oriented data
   b) Text, files containing data
   c) Data in the form of audio or video
   d) All of the above

8. Column header is refered to as .....
   a) Table
   b) Relation
   c) Attributes
   d) Domain

9. Queries to a database .....
   a) Are written in English
   b) Can use aggregate functions like SUM and COUNT
   c) Both a. and b.
   d) None of the above

10. Which of the following is not an example of DBMS?
   a) MySQL
   b) Microsoft Acess
   c) IBM DB2
   d) Google

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11. Which language has recently become the de-facto standard for interfacing application programs with relational database system?
a) Oracle
b) SQL
c) 4GL
d) None of the above

12. What is the purpose of indexing in a database?
a) To store data in a hierarchical structure
b) To organize data in a sequential manner
c) To speed up data retrieval operations
d) To encrypt sensitive data

13. Which of the following is not a type of database constraint?
a) Primary Key constraint
b) Foreign Key constraint
c) Mandatory constraint
d) Optional constraint

14. In a relational database, what is a primary key?
a) A key that uniquely identifies each record in a table
b) A key that is used for foreign key references
c) A key that is shared among multiple tables
d) A key that is randomly generated

15. In SQL, which command is used to add new rows to a table?
a) ALTER TABLE
b) ADD ROW
c) INSERT
d) APPEND

16. It is more accurate to define a _____ as a variety of different record types that are treated as a single unit
a) Database
b) Table
c) Record
d) Flat File

17. Which of the following is a function of the DBMS?
a) Storing data
b) Providing multi-users access control
c) Data Integrity
d) All of the above

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18.Which of the following is known as a set of entities of the same type that share same properties, or attributes?
a) Relation set
b) Tuples
c) Entity set
d) Entity Relation model

19.Disadvantages of File systems to store data are .....
a) Data redundancy and inconsistency
b) Difficulty in accessing data
c) Data isolation
d) All of the above

20.What does an RDBMS consist of?
a) Collection of Records
b) Collection of Keys
c) Collection of Tables
d) Collection of Fields

21.Which of the following is a database administrator's function?
a) Database design
b) Backing up the database
c) Performance monitoring
d) All of the above

22.A _____ contains fields of data about one entity.
a) Record
b) File
c) Fields
d) Document

23. What is the purpose of a foreign key in a relational database?
a) To uniquely identify each record in a table
b) To establish a link between two tables
c) To enforce data integrity constraints
d) To sort data in ascending order

24.Which SQL Query is used to remove a table and all its data from the database?
a) Drop table
b) Delete
c) Purge
d) Remove

25.Which of the following options is used for retrieval of data?
a) Stack
b) Data Structure
c) Linked list
d) Query

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SECTION B

[75 Marks]

*Answer any **three** questions in this section.*

1. a. Explain the concept of database normalization in the context of a relational DBMS. [10 Marks]
   b. What are the 3 primary benefits of normalization? [15 Marks]

2. Discuss the role of indexes in a database system and explain how they contribute to improving query performance. Provide examples to support your explanation. [25 Marks]

3. (a) Differentiate between Database and DBMS [10 marks] (b) Explain the following terms Database. [15marks] i. Record,
   ii. attribute,
   iii. datatype,

4. Explain *five* challenges of Database Management System (DBMS). [25marks]

5. a. Differentiate between Data and Information [5 marks]

   b. Explain the role involved in the DBMS environment. [20 marks]

- Data And Database Administrators
- Database Designers
- Application Developers
- End-Users

Good Luck